

The Rise of Second-Hand Intelligence:

How Generative AI Is Reshaping Human Expertise
and Original Thought

MBAI Program — AI Programming Research Paper

Comprehensive Research, Methodology & Implementation Guide

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Paper Title	The Rise of Second-Hand Intelligence
Paper Type	Hybrid Conceptual-Computational Research
Methodology	Theoretical Analysis + NLP + Machine Learning
Dataset	3 Categories: Human, AI-Generated, Human-Edited AI
Core Construct	Second-Hand Intelligence (novel operationalization)
Target Venue	Computers & Education / AI & Society / AIED

Abstract: This research plan provides a comprehensive blueprint for a graduate-level interdisciplinary paper examining how generative AI is transforming human expertise, originality, and independent cognition. The plan covers theoretical framing, literature strategy, a full NLP pipeline, machine learning methodology, visualization guidance, and practical implementation advice — designed to produce a publication-quality, computationally grounded academic paper.

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PART I — THEORETICAL FRAMING AND ACADEMIC POSITIONING

1.1 Core Theoretical Contribution

Your central construct — **"second-hand intelligence"** — is novel enough to anchor a real contribution. Frame it precisely: second-hand intelligence is not AI use per se, but a *cognitive disposition* in which the agent presents, refines, or acts upon machine-generated cognition without having generated the underlying epistemic work. The distinction matters because it separates tool use (benign) from epistemic substitution (the problem your paper targets).

Position the paper at the intersection of three fields: **Cognitive Science** (extended cognition, cognitive offloading), **Science and Technology Studies** (techno-epistemology, human-machine knowledge co-production), and **AI/HCI** (human-AI collaboration, AI-mediated reasoning). This triangulation gives the paper its interdisciplinary edge and makes it harder to dismiss as either purely philosophical or purely technical.

1.2 Recommended Theoretical Frameworks

Primary Frameworks

Framework	Application to Your Paper
Extended Mind Thesis (Clark & Chalmers, 1998)	Argues that cognition extends beyond the skull into tools and environments. Draw on this and critique its limits: classical extended mind theory assumes the agent remains the epistemic initiator. Generative AI complicates this because the external system is now generative, not merely storage.
Cognitive Offloading (Risko & Gilbert, 2016)	Distinguishes motor offloading from cognitive offloading. Key to your paper: distinguish productive offloading (outsourcing mechanical tasks, freeing higher cognition) from degenerative offloading (outsourcing the higher cognition itself).
Distributed Cognition (Hutchins, 1995)	Cognitive systems are distributed across agents and artifacts. Useful for framing human-AI collaboration positively, but your paper should show that distributed cognition implies interdependence, not autonomy — setting up your epistemic dependence argument.
Epistemic Autonomy (Goldman, 1999; Fricker, 2007)	From social epistemology. Fricker's epistemic injustice framework is adjacent but relevant; your concern is epistemic atrophy — a new contribution you can define and operationalize.
Technology as Pharmacology (Stiegler, 2010)	Technology is simultaneously a poison and a cure. This is your philosophical backbone for the tension between augmentation and overdependence — one of the most theoretically rich positions for your argument.
Expertise Reversal Effect (Kalyuga, 2007)	From educational psychology. As AI provides scaffolding that expert cognition no longer needs to build independently, performance may improve short-term while underlying competence atrophies. Directly applicable to your expertise transformation argument.

Human-AI Collaboration (Licklider, 1960; Shneiderman, 2022)	Licklider's man-computer symbiosis is the founding text of human-computer augmentation theory. Shneiderman's recent work on human-centered AI updates this for the generative era. Use both to establish your historical arc.
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Secondary / Supporting Frameworks

- **Flow and Deep Work** (Csikszentmihalyi, 1990; Newport, 2016) — For the argument that AI-mediated cognition may undermine conditions necessary for intellectual depth.
- **Threshold Concepts** (Meyer & Land, 2003) — Certain forms of expertise require struggling through difficulty; offloading may prevent threshold crossing.
- **Zone of Proximal Development** (Vygotsky, 1978) — AI as scaffolding can either support development or permanently substitute for it.
- **Automation Complacency** (Parasuraman & Manzey, 2010) — Strong empirical basis for cognitive skill degradation under automation reliance.

1.3 Measurable Hypotheses

From your research questions, derive these testable hypotheses to anchor the computational section:

H1 — Originality	AI-generated texts will exhibit lower lexical diversity, lower semantic range, and higher intertextual similarity to one another compared to human-written texts on identical prompts.
H2 — Homogenization	Texts across different AI-generated responses will be more similar to each other than texts across different human-written responses, suggesting structural convergence.
H3 — Hybrid Attenuation	Human-edited AI texts will show features intermediate between pure human and pure AI text, but cluster closer to AI on most structural metrics.
H4 — Detectability	Machine learning classifiers will achieve meaningful accuracy distinguishing the three text categories, with precision varying by feature set.
H5 — Expertise Markers	NLP measures of reasoning depth (argument markers, hedging, lexical sophistication, sentence complexity) will differ significantly across categories.

PART II — LITERATURE STRATEGY

2.1 How Many Sources You Need

For a strong graduate-level research paper with a computational component: **55–80 sources** is the target. Fewer than 55 will feel thin; more than 80 without a clear taxonomy will feel unfocused.

Source Category	Count	Purpose
Foundational theoretical works	8–12	Non-negotiable theoretical anchors
Core empirical studies	15–20	Recent peer-reviewed directly relevant work
Methodological / computational references	10–15	NLP methods, ML models, text analysis
Contextual and contemporary sources	12–18	AI policy, education, professional implications
Seminal historical texts	5–8	Used sparingly but deliberately

2.2 Literature Categories

Thematic Category	Sources	Maps to Section
Cognitive offloading & extended cognition	10–12	Literature Review — theoretical spine
Human-AI collaboration & HCI	8–10	Conceptual Framework
AI-mediated cognition & epistemic dependence	8–10	Core novel argument
Expertise, skill acquisition, learning	8–10	Expertise transformation
Originality, creativity, authorship	6–8	Originality section
NLP methods & computational text analysis	10–12	Methodology justification
AI detection & text classification	6–8	ML methodology
Academic integrity & AI in education	6–8	Implications section
Generative AI capabilities & limitations	5–6	Background / Context

2.3 Seminal Papers — Read These First

Cognitive Science

- Clark, A., & Chalmers, D. (1998). The Extended Mind. *Analysis*, 58(1), 7–19.
- Risko, E. F., & Gilbert, S. J. (2016). Cognitive Offloading. *Trends in Cognitive Sciences*, 20(9), 676–688.
- Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

- Sparrow, B., Liu, J., & Wegner, D. M. (2011). Google Effects on Memory. *Science*, 333(6043), 776–778. (*Your most relevant empirical precursor*)

AI & HCI

- Licklider, J. C. R. (1960). Man-Computer Symbiosis. *IRE Transactions on Human Factors in Electronics*.
- Shneiderman, B. (2022). *Human-Centered AI*. Oxford University Press.
- Brynjolfsson, E., & McAfee, A. (2014). *The Second Machine Age*. W.W. Norton.

NLP / Computational

- Devlin, J., et al. (2019). BERT: Pre-training of Deep Bidirectional Transformers. *NAACL-HLT*.
- Guo, B., et al. (2023). How Close is ChatGPT to Human Experts? *arXiv:2301.07597*.
- Liang, W., et al. (2024). Monitoring AI-Modified Content at Scale. *PNAS*. (*AI homogenization evidence — extremely relevant*)
- Jawahar, G., et al. (2020). Automatic Detection of Machine Generated Text. *COLING*.

2.4 Database Search Strategy

Database	Best For	Key Search Strings
Google Scholar	Broad discovery, citing papers	"cognitive offloading" AND "artificial intelligence" "AI-generated text" AND "originality"
Scopus / WoS	Systematic citation-verified search	TITLE-ABS-KEY("generative AI" AND "cognitive") filter: 2018–2026, peer-reviewed
ACM Digital Library	HCI and computing papers	"AI writing assistance" cognitive "human-AI" collaboration expertise
IEEE Xplore	ML/NLP technical methodology	"AI-generated text detection" "NLP text classification" "transformer authorship"
ScienceDirect	Cognitive science and psychology	Journals: Computers in Human Behavior, Cognition, Trends in Cognitive Sciences

2.5 High-Value Keyword Combinations

- "generative AI" AND "intellectual autonomy"
- "LLM" AND "cognitive offloading"
- "AI writing" AND "creativity" AND "originality"
- "second-hand knowledge" OR "vicarious cognition" AND AI
- "AI text detection" AND "machine learning"
- "epistemic dependence" AND "technology"
- "automation" AND "skill atrophy" OR "skill degradation"
- "human-AI collaboration" AND "deep thinking" OR "critical thinking"
- "ChatGPT" AND "academic writing" AND "authenticity"

- "AI homogenization" AND "writing style"

PART III — METHODOLOGY DESIGN

3.1 Hybrid Methodology Architecture

Structure your methodology as two interconnected phases:

- **Phase 1 — Conceptual Analysis:** Systematic review and theoretical synthesis using your frameworks. This produces your conceptual model of second-hand intelligence.
- **Phase 2 — Computational Analysis:** NLP + ML analysis of text samples that operationalizes and empirically tests key claims from Phase 1.

The bridge between phases is operationalization: theoretical claims (e.g., 'AI text reduces epistemic originality') become measurable proxies (e.g., lexical diversity, semantic similarity, readability scores, classifier confidence). Every computational finding should speak back to the theory.

3.2 Dataset Design

Recommended prompts (use 3–5, not just 1):

- Does generative AI strengthen or weaken independent thinking?
- What does it mean to be an expert in the age of AI?
- Is AI-generated writing genuinely creative?
- How should universities respond to AI-assisted academic work?
- What is lost when humans delegate thinking to machines?

These opinion/analytical tasks maximize variation between human and AI responses and directly mirror your theoretical argument.

Category	Description	Sample Target	Key Notes
A — Human-Written	Genuine human responses from colleagues or written across multiple sessions	6–8 per prompt	Use informed consent; anonymize data
B — AI-Generated	Pure ChatGPT-4o / Claude outputs with no editing	6–8 per prompt	Log model name and version for reproducibility
C — Human-Edited AI	AI outputs substantially revised with personal examples and restructured arguments	6–8 per prompt	Most theoretically interesting category

Recommended CSV schema: **id, prompt_id, category, text, word_count, source_model**

PART IV — NLP PIPELINE AND CODING WORKFLOW

4.1 Step 1: Environment Setup

```
pip install pandas numpy scikit-learn matplotlib seaborn
pip install nltk spacy textblob textstat
pip install sentence-transformers # semantic similarity
pip install umap-learn           # embedding visualization
python -m spacy download en_core_web_sm
```

4.2 Step 2: Data Ingestion and Preprocessing

```
import pandas as pd, nltk, re
from nltk.tokenize import word_tokenize, sent_tokenize
from nltk.corpus import stopwords

nltk.download(['punkt', 'stopwords', 'averaged_perceptron_tagger'])

def preprocess(text):
    text = text.lower().strip()
    tokens = word_tokenize(text)
    tokens_clean = [t for t in tokens if t.isalpha()]
    return tokens, tokens_clean # keep both raw and clean
```

4.3 Step 3: Feature Extraction

Feature extraction is the most theoretically important step. Each feature is a computational proxy for a theoretical construct in your paper.

Lexical Diversity

```
def type_token_ratio(tokens):
    return len(set(tokens)) / len(tokens) if tokens else 0

def hapax_legomena_ratio(tokens):
    freq = nltk.FreqDist(tokens)
    hapax = [w for w, c in freq.items() if c == 1]
    return len(hapax) / len(tokens) if tokens else 0
```

Readability

```
import textstat
features['flesch_reading_ease'] = textstat.flesch_reading_ease(text)
features['gunning_fog']        = textstat.gunning_fog(text)
features['flesch_kincaid_grade'] = textstat.flesch_kincaid_grade(text)
features['smog_index']          = textstat.smog_index(text)
```

Hedging & Epistemic Voice (key for originality argument)

```
hedging_words = ['perhaps', 'possibly', 'might', 'may', 'seems', 'appears',
                 'arguably', 'likely', 'uncertain', 'suggest', 'indicate',
                 'believe', 'think', 'consider', 'wonder']
first_person = ['i', 'my', 'me', 'myself', 'we', 'our']
```

```
features['hedging_ratio'] = sum(1 for t in tokens if t in hedging_words) / len(tokens)
features['first_person_ratio'] = sum(1 for t in tokens if t in first_person) / len(tokens)
```

Within-Category Semantic Similarity (homogenization test)

```
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

model = SentenceTransformer('all-MiniLM-L6-v2')
embeddings = model.encode(df['text'].tolist())

ai_sim = cosine_similarity(embeddings[df['category']=='AI']).mean()
human_sim = cosine_similarity(embeddings[df['category']=='Human']).mean()
# If ai_sim >> human_sim -> direct evidence of AI text homogenization
```

This within-category similarity comparison is your most powerful finding. If AI texts are more similar to each other than human texts are to each other, you have direct empirical evidence for homogenization — the strongest result in the paper.

4.4 Step 4: Machine Learning Classification

```
from sklearn.model_selection import cross_val_score, StratifiedKFold
from sklearn.linear_model import LogisticRegression
from sklearn.naive_bayes import MultinomialNB
from sklearn.ensemble import RandomForestClassifier
from sklearn.svm import SVC
from sklearn.metrics import classification_report

models = {
    'Logistic Regression': LogisticRegression(max_iter=1000),
    'Naive Bayes': MultinomialNB(),
    'Random Forest': RandomForestClassifier(n_estimators=100, random_state=42),
    'SVM': SVC(kernel='linear', probability=True)
}

for name, clf in models.items():
    cv = cross_val_score(clf, X, y, cv=StratifiedKFold(5), scoring='f1_macro')
    print(f"{name}: {cv.mean():.3f} +/- {cv.std():.3f}")
```

Use 5-fold stratified cross-validation as your primary evaluation method given small sample sizes. Report F1-macro (handles class imbalance). This is more robust and publishable than a single train/test split.

PART V — VISUALIZATIONS

5.1 Recommended Figures (6–8 Total)

Figure	Type	What It Shows	Priority
Figure 1	Conceptual Diagram	Framework: cognitive offloading → second-hand intelligence → epistemic dependence	Essential
Figure 2	Boxplots	Key NLP metrics (TTR, readability, hedging, sentence length) by category	Essential
Figure 3	Similarity Heatmaps	Within-category cosine similarity matrices — 3 panels (AI, Human, Hybrid). Shows homogenization.	Essential
Figure 4	UMAP/PCA Scatter	2D BERT embedding space colored by category. If clusters are distinct, validates ML approach.	High
Figure 5	Model Comparison Bar	F1 scores per model with cross-validation error bars	High
Figure 6	Confusion Matrix	Best model confusion matrix — which categories are hardest to distinguish?	High
Figure 7	Feature Importance Plot	Top 15–20 discriminating features ranked by importance, annotated by type	High
Table 1	Descriptive Statistics	Mean ± SD for all NLP features per category	Essential
Table 2	Classification Results	Precision, Recall, F1 per class per model — standard ML reporting format	Essential

5.2 UMAP Visualization Code

```
from umap import UMAP
import matplotlib.pyplot as plt

reducer = UMAP(n_components=2, random_state=42)
embedding_2d = reducer.fit_transform(embeddings)

plt.figure(figsize=(8, 6))
colors = {'Human': '#2196F3', 'AI': '#F44336', 'Hybrid': '#4CAF50'}
for cat, color in colors.items():
    mask = df['category'] == cat
    plt.scatter(embedding_2d[mask, 0], embedding_2d[mask, 1],
                label=cat, color=color, alpha=0.7, s=60)
plt.legend(fontsize=11)
plt.title("BERT Embedding Space by Text Category", fontsize=13)
```

```
plt.tight_layout()  
plt.savefig("figures/umap_embeddings.png", dpi=300, bbox_inches='tight')
```

PART VI — MEASURING ORIGINALITY AND SECOND-HAND INTELLIGENCE

6.1 Originality Proxy Score (OPS)

Propose a composite metric as a novel operationalization of 'originality'. This gives you a single quantifiable construct you can compare across categories:

```
# OPS = weighted combination of originality-proxying features
# Weights derived from classifier feature importances

OPS = (w1 * TTR
      + w2 * hapax_ratio
      + w3 * first_person_ratio
      + w4 * hedging_density
      + w5 * (1 - cosine_similarity_to_category_centroid))
```

Frame this explicitly as a proposed proxy metric requiring future validation — this is academically honest and appropriate for a course paper with a small sample. The act of proposing and testing a new operationalization is itself a contribution.

6.2 LLM-as-Evaluator (Optional but Strong)

Use a structured GPT-4 prompt to score each text on four dimensions. Document your evaluation prompt exactly and include it in an appendix.

Dimension	Scoring	Theoretical Link
Originality	1–5	Does the text offer a perspective not predictable from the prompt alone?
Reasoning Depth	1–5	Are claims supported with developed, multi-step arguments?
Critical Engagement	1–5	Does the text question assumptions or consider counterarguments?
Authorial Voice	1–5	Is there a distinct, consistent perspective throughout?

Acknowledge in your limitations section that GPT-4 may rate its own outputs favorably — this is a validity threat worth naming explicitly.

PART VII — COMMON MISTAKES TO AVOID

7.1 Structural Mistakes

- Letting the code section feel like an appendix rather than an integral part of the argument. Every computational finding must be explicitly interpreted through your theoretical framework.
- Writing a theory section that never gets revisited in the Discussion. Your Discussion must explicitly revisit each framework (extended mind, cognitive offloading, etc.) in light of your findings.
- Treating your small sample as more definitive than it is. Frame as 'proof of concept / exploratory empirical validation' — honest and defensible.

7.2 Methodological Mistakes

- Using only accuracy as your evaluation metric — use F1-macro with cross-validation.
- Ignoring prompt effects — analyze texts within-prompt before across-prompt.
- Treating the three categories as perfectly clean — they aren't; acknowledge this in limitations.
- Not normalizing features before ML — critical for SVM and Logistic Regression.

7.3 Writing and Framing Mistakes

- Defining AI without precision — specify which systems (instruction-tuned LLMs, GPT-4 class) and when your observations apply.
- Conflating correlation with causation. Your data shows differences, not mechanisms.
- Over-claiming: 'AI weakens human thinking' is not proven by your data. 'AI-generated texts show measurable differences in linguistic markers associated with originality, and these differences are reliably detectable by ML classifiers' is what your data can support.

PART VIII — WHAT WILL IMPRESS YOUR PROFESSOR MOST

Rank	Element	Why It Impresses
1	Conceptual Framework Diagram	A clear, original visual model synthesizing your theoretical position. Professors remember visual arguments.
2	Within-Category Similarity Analysis	Empirical, direct, and novel. If AI texts cluster tighter than human texts, you have a memorable result.
3	Originality Proxy Score (OPS)	Proposing a new operationalized metric shows theoretical-empirical integration — rare in course papers.
4	LLM-as-Evaluator section	Using an AI to evaluate AI-assisted cognition has elegant irony and demonstrates technical sophistication.
5	Cross-Validation Rigor	StratifiedKFold + multiple models + F1-macro signals genuine ML literacy, not tutorial-level work.
6	Theoretical Precision	Clearly distinguishing second-hand intelligence from mere AI use and from cognitive offloading shows conceptual maturity.

PART IX — ETHICAL CONSIDERATIONS AND LIMITATIONS

9.1 Ethical Considerations

- **Informed consent:** Participants must know their writing will be analyzed and classified. Data should be anonymized.
- **AI model transparency:** Document which model versions generated your AI samples — outputs change with version, affecting reproducibility.
- **Classification risks:** A classifier that labels human writing as 'AI' could be weaponized in academic misconduct cases. Discuss this misuse potential explicitly.
- **Normative framing:** The paper should avoid implying AI use is categorically unethical. The argument is about how AI is used, not whether it should be.
- **Bias in AI evaluators:** If using GPT-4 to score originality, it may rate its own outputs favorably — acknowledge as a validity threat.

9.2 Limitations to Address

- Small, convenience sample — frame as exploratory/proof-of-concept
- Single-language analysis (English only)
- Prompts focused on AI discourse — domain-specific, may not generalize

- Binary categories underrepresent the complexity of real AI-assisted writing
- Cross-validation on small N still has high variance
- 'Originality' as a construct is contested — your proxy measures are proxies, not ground truth

PART X — PROJECT ORGANIZATION

10.1 Recommended GitHub / Folder Structure

```
second-hand-intelligence/  
|-- data/  
| |-- raw/          # original text files  
| |-- processed/    # cleaned CSV  
| +-- README.md     # data provenance documentation  
|-- notebooks/  
| |-- 01_preprocessing.ipynb  
| |-- 02_feature_extraction.ipynb  
| |-- 03_ml_classification.ipynb  
| +-- 04_visualization.ipynb  
|-- src/  
| |-- features.py  
| |-- models.py  
| +-- utils.py  
|-- outputs/  
| |-- figures/  
| +-- tables/  
|-- requirements.txt  
+-- README.md
```

Use Jupyter notebooks for the paper — they keep code, output, and narrative together, and your professor can reproduce results exactly. Export cleaned notebooks as PDFs or include them as appendix.

PART XI — PUBLICATION POSITIONING

11.1 Target Venues

Venue	Type	Fit
Computers & Education (Elsevier)	Journal	Strong fit — accepts hybrid conceptual-empirical work on AI in learning contexts
Computers in Human Behavior (Elsevier)	Journal	HCI + cognitive science — directly covers your theoretical territory
AI & Society (Springer)	Journal	Explicitly interdisciplinary; accepts philosophical-technical arguments
Journal of Information Technology	Journal	Professional and academic AI implications; accessible scope
First Monday	Open Access	Forward-thinking digital scholarship; open access increases reach

ACM CHI	Conference	If you strengthen the HCI angle post-course
AIED (AI in Education)	Conference	Perfect topical fit for the educational implications angle

11.2 What Would Need to Happen Before Submission

- Increase sample size to 60+ responses per category
- Conduct inter-rater reliability testing on the LLM evaluation scores
- Add one quantitative study from secondary data (e.g., analysing AI-assisted writing on public platforms)
- Expand to a second language or domain to test generalizability
- Validate the Originality Proxy Score against human expert ratings

STRATEGIC STARTING ADVICE

The three things most likely to determine the quality of the final paper are:

- **Dataset quality:** Give yourself more time than you expect to collect clean human-written samples. This is the single biggest practical bottleneck.
- **Theory-findings integration:** After running your analysis, spend equal time interpreting each computational result through your theoretical frameworks. Don't let the code section and the theory section talk past each other.
- **Epistemically honest framing:** Your sample is small and your metrics are proxies. Say so clearly and frame your study as exploratory. This is not a weakness — it is intellectual maturity, and it is exactly what reviewers and professors respect.

Recommended first action: Build your dataset and run the within-category semantic similarity analysis first. If AI texts are measurably more homogeneous than human texts, that single finding becomes the empirical anchor for everything else — and the rest of the analysis, Discussion, and theoretical interpretation builds outward from it.

Research Plan compiled June 2026 | MBAI Program — AI Programming | Quayci Clement-Yorke